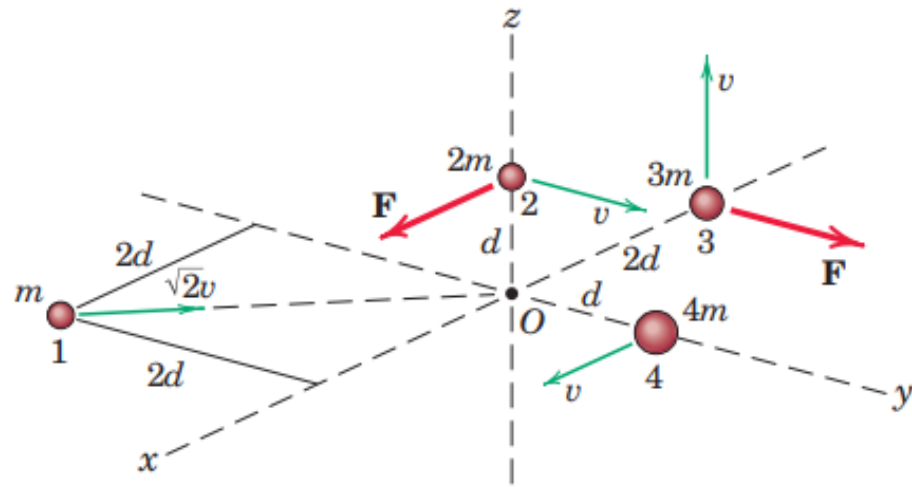


Kinetics of System of Particles

Dr. Atanu Banerjee

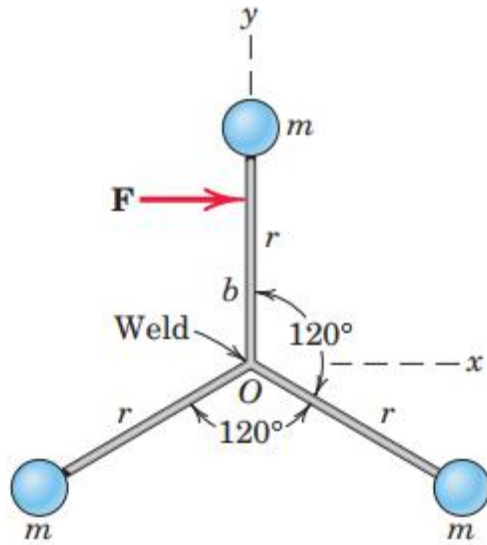
Example-1

The system of four particles has the indicated particle masses, positions, velocities, and external forces. Determine $\bar{\mathbf{r}}$, $\dot{\bar{\mathbf{r}}}$, $\ddot{\bar{\mathbf{r}}}$, T , $\mathbf{G}$, $\mathbf{H}_O$, $\dot{\mathbf{H}}_O$, $\mathbf{H}_G$, and $\dot{\mathbf{H}}_G$.



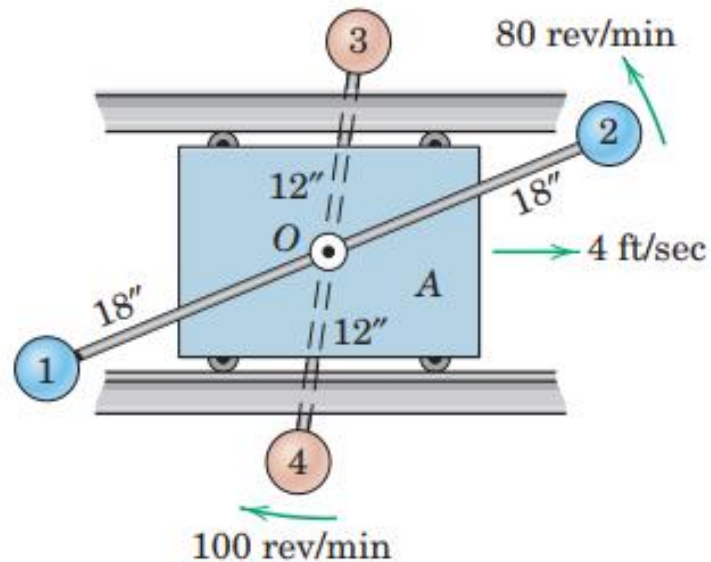
Example-2

Each of the three balls has a mass m and is welded to the rigid equiangular frame of negligible mass. The assembly rests on a smooth horizontal surface. If a force $\mathbf{F}$ is suddenly applied to one bar as shown, determine (a) the acceleration of point O and (b) the angular acceleration $\ddot{\theta}$ of the frame.



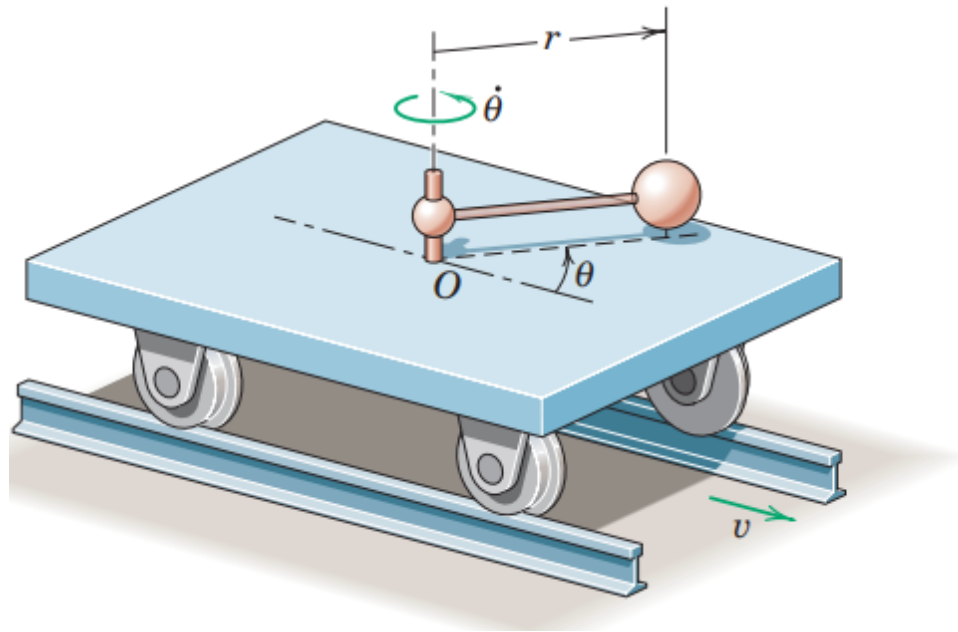
Example-3

The 32.2-lb carriage A moves horizontally in its guide with a speed of 4 ft/sec and carries two assemblies of balls and light rods which rotate about a shaft at O in the carriage. Each of the four balls weighs 3.22 lb. The assembly on the front face rotates counterclockwise at a speed of 80 rev/min, and the assembly on the back side rotates clockwise at a speed of 100 rev/min. For the entire system, calculate (a) the kinetic energy T , (b) the magnitude G of the linear momentum, and (c) the magnitude H_O of the angular momentum about point O .



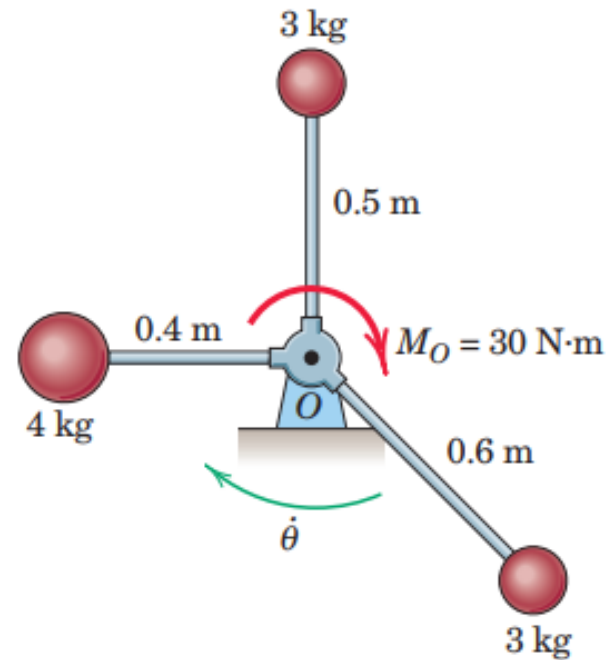
Example-4

The small car, which has a mass of 20 kg, rolls freely on the horizontal track and carries the 5-kg sphere mounted on the light rotating rod with $r = 0.4$ m. A geared motor drive maintains a constant angular speed $\dot{\theta} = 4$ rad/s of the rod. If the car has a velocity $v = 0.6$ m/s when $\theta = 0$, calculate v when $\theta = 60^\circ$. Neglect the mass of the wheels and any friction.



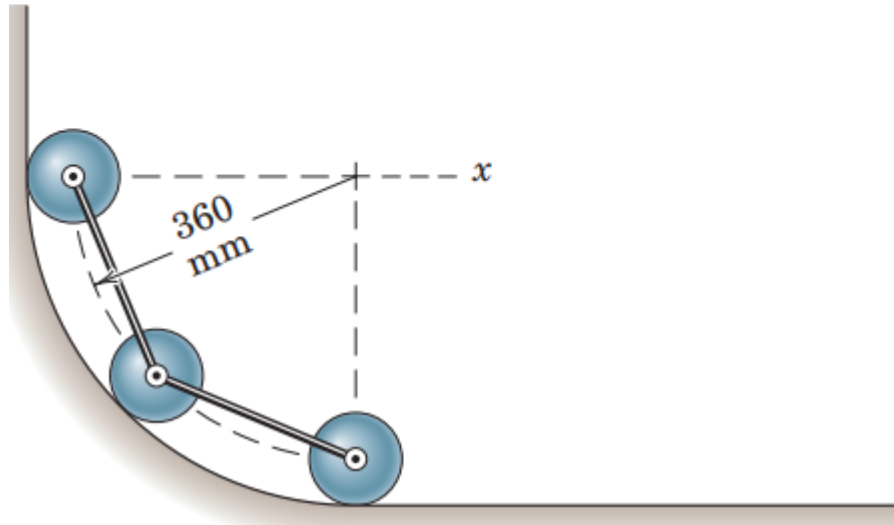
Example-5

The three small spheres are welded to the light rigid frame which is rotating in a horizontal plane about a vertical axis through O with an angular velocity $\dot{\theta} = 20 \text{ rad/s}$. If a couple $M_O = 30 \text{ N}\cdot\text{m}$ is applied to the frame for 5 seconds, compute the new angular velocity $\dot{\theta}'$.



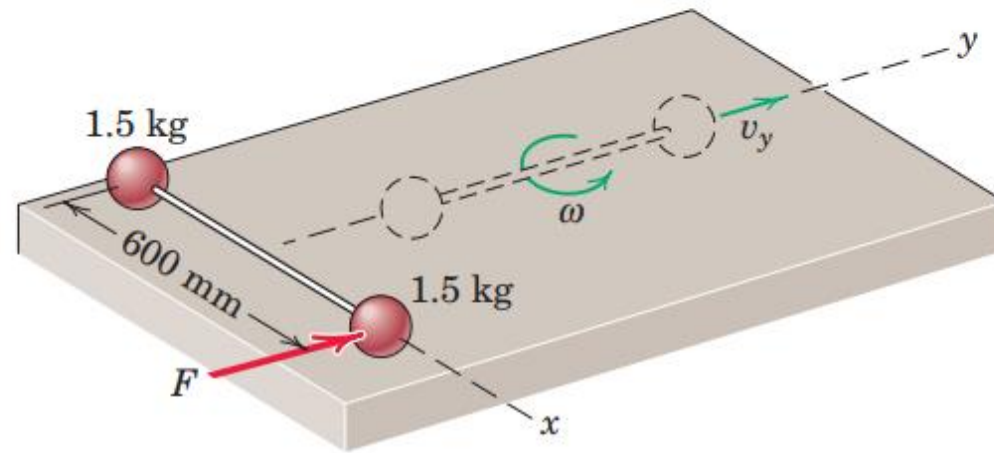
Example-6

The three small steel balls, each of mass 2.75 kg, are connected by the hinged links of negligible mass and equal length. They are released from rest in the positions shown and slide down the quarter-circular guide in the vertical plane. When the upper sphere reaches the bottom position, the spheres have a horizontal velocity of 1.560 m/s. Calculate the energy loss ΔQ due to friction and the total impulse I_x on the system of three spheres during this interval.



Example-7

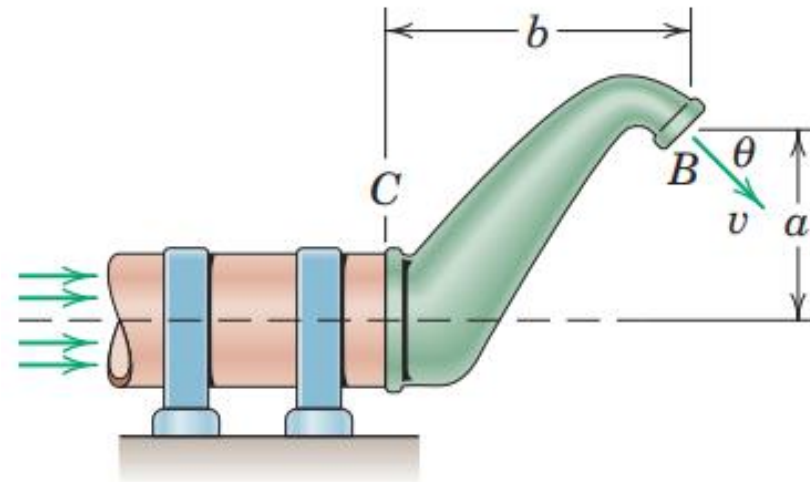
The two spheres are rigidly connected to the rod of negligible mass and are initially at rest on the smooth horizontal surface. A force F is suddenly applied to one sphere in the y -direction and imparts an impulse of $10 \text{ N}\cdot\text{s}$ during a negligibly short period of time. As the spheres pass the dashed position, calculate the velocity of each one.



Example-8

System with Steady Mass Flow

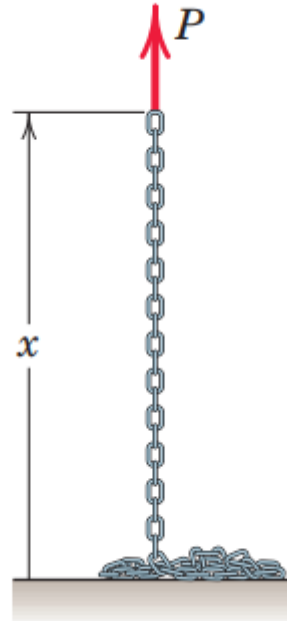
The offset nozzle has a discharge area A at B and an inlet area A_0 at C . A liquid enters the nozzle at a static gage pressure p through the fixed pipe and issues from the nozzle with a velocity v in the direction shown. If the constant density of the liquid is ρ , write expressions for the tension T , shear Q , and bending moment M in the pipe at C .



Example-9

System with varying mass

The end of a chain of length L and mass ρ per unit length which is piled on a platform is lifted vertically with a constant velocity v by a variable force P . Find P as a function of the height x of the end above the platform. Also find the energy lost during the lifting of the chain.



Example-10

Fresh water issues from the two 30-mm-diameter holes in the bucket with a velocity of 2.5 m/s in the directions shown. Calculate the force P required to give the bucket an upward acceleration of 0.5 m/s^2 from rest if it contains 20 kg of water at that time. The empty bucket has a mass of 0.6 kg.

